

## PART-1- numberToChar Function

This function takes the number of characters in the comment line as integers and then converts them to characters.(For this reason I added '0').

```
1 #include <stdio.h>
2
3
4 void numberToChar(int charCount , int myKey){ /* turn numbers to char in the comment line. */
5     char alphabet[61] = {'a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'j', 'k',
6         'l', 'm', 'n', 'o', 'p', 'q', 'r', 's', 't', 'u', 'v', 'w', 'x', 'y', 'z',
7         '(', '<', '=', '+', ')', '[', '*', '/', ']', '{', '>', '!', '-', '}', '?',
8         '\\', '&', '|', '%', '_', ':', '"', '#', '.', '\'', '0', '1', '2', '3', '4',
9         '5', '6', '7', '8', '9'};
10
11     int firstDigit = charCount % 10; /* take mod to find first digit. */
12     int secondDigit = (charCount - firstDigit) / 10 ; /* find second digit. */
13     char firstDigitChar, secondDigitChar;
14     firstDigitChar = firstDigit + '0'; /* convert number to char according to alphabet to compare */
15     secondDigitChar = secondDigit + '0';
16
17     int i=0;
18     for(i ; i <= 61 ; i++){
19         int value = i + myKey;
20         if(alphabet[i] == secondDigitChar){
21             printf("%c", alphabet[value % 61]); /* print the matches digits to console*/
22             secondDigitChar += '0';
23         }
24         if(alphabet[i] == firstDigitChar){
25             printf("%c", alphabet[value % 61]);
26             firstDigitChar += '0';
27         }
28     }
29
30 int main(){
```

## PART-1-MAIN-1

```
29 int main(){
30
31     char alphabet[61] = {'a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'j', 'k',
32         'l', 'm', 'n', 'o', 'p', 'q', 'r', 's', 't', 'u', 'v', 'w', 'x', 'y', 'z',
33         '(', '<', '=', '+', ')', '[', '*', '/', ']', '{', '>', '!', '-', '}', '?',
34         '\\', '&', '|', '%', '_', ':', '"', '#', '.', '\'', '0', '1', '2', '3', '4',
35         '5', '6', '7', '8', '9'};
36
37     long long int studentId = 230104004174;
38     int digitSums = 0;
39     int myKey;
40
41     while(1){ /* find key according to student ID.*/
42         long long int tempStudentId = studentId;
43         while (tempStudentId > 0){
44
45             digitSums += tempStudentId % 10;
46             tempStudentId /= 10;
47         }
48         if(digitSums < 10){ /* continue until find the one digit number. */
49             break;
50         }
51         studentId = digitSums;
52         digitSums = 0;
53     }
54
55     myKey = digitSums;
56     char tempChar;
57     int isCommentFlag = 0;
58     int charCount = -1; /* due to structre of the sentence , I started from -1. */
59 }
```

## PART-1-MAIN-2

```
--
60 while(scanf("%c", &tempChar) != EOF){
61     if(tempChar == ' '){ /* print spaces and end of the lines.*/
62         printf("%c", ' ');
63     }else if(tempChar == '\n'){
64         printf("%c", '\n');
65     }
66
67     if(tempChar == '/'){ /* check if comment line start or not */
68         char commentCh;
69         scanf("%c", &commentCh);
70         if(commentCh == '*'){
71             isCommentFlag = 1;
72         }
73     }else if(tempChar == '*'){ /* check if comment line finish or not */
74         char commentCh2;
75         scanf("%c", &commentCh2);
76         if(commentCh2 == '/'){
77             printf("@ ");
78             numberToChar(charCount, myKey);
79             isCommentFlag = 0;
80         }
81     }
82
83     if(isCommentFlag == 0){
84         int i=0;
85         for(i ; i <= 61 ; i++){
86             int value = i + myKey;
87             if(alphabet[i] == tempChar && tempChar != '*'){
88                 printf("%c", alphabet[value % 61]);
89             }
90         }
91     }else{
92         if(tempChar != ' '){
93             charCount += 1;
94         }
95     }
96
97 }
98
99
100 }
101 return 0;
102 }
```

## ENCRYPTED FILE (OUTPUT-1)

Open [icon] encrypted.txt  
~/Desktop

```
1 5qvkt=lm {( <lqw6p%
2
3 qv< uiqv]- | @ aa
4 qv< v=ujmz2wvm > b3
5 qv< v=ujmz2<)w > c3
6 qv< ill > v=ujmz2wvm ! v=ujmz2<)w3
7 zm<=zv 83
8 "|
```

## PART-2-MAIN-1

```
1 #include <stdio.h>
2 int main(){
3
4     char alphabet[61] = {'a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'j', 'k', 'l', 'm', 'n', 'o', 'p', 'q', 'r', 's', 't', 'u', 'v', 'w', 'x', 'y', 'z', '(', '<', '=', '+', ')', '[', '*', '/',
5     ']', '!', '>', '!', '-', ']', '!', '\\', '8', '!', '%', '!', '_', '!', '+', '#', '!', '\\', '0', '1', '2', '3', '4', '5', '6', '7', '8', '9'};
6
7     long long int studentId = 230104004174;
8     int digitsSums = 0;
9     int nyKey;
10
11     while(1){ /* find key according to student ID.*/
12         long long int tempStudentId = studentId;
13         while (tempStudentId > 0){
14             digitSums += tempStudentId % 10;
15             tempStudentId /= 10;
16         }
17         if(digitSums < 10){ /* continue until find the one digit number. */
18             break;
19         }
20         studentId = digitSums;
21         digitSums = 0;
22     }
23
24     nyKey = digitSums;
25     char tempChar;
26 }
```

## PART-2-MAIN-2

```
26
27     while(scanf("%c", &tempChar) != EOF){
28         int i=0;
29         if(tempChar == ' '){
30             printf("%c", ' ');
31         }else if(tempChar == '\\n'){
32             printf("%c", '\\n');
33         }
34         if(tempChar == '@'){ /* check if comment line start or not and decrypted it. */
35             printf("/*There is a */");
36             char tempCharForCommentLine;
37             for(int k = 0; k<3; k++){
38                 scanf("%c", &tempCharForCommentLine);
39                 int i = 0;
40                 for(i; i <= 61; i++){
41                     int value = i - nyKey;
42                     if(value < 0){
43                         value += 61;
44                     }
45                     if(alphabet[i] == tempCharForCommentLine){
46                         printf("%c", alphabet[value]);
47                     }
48                 }
49             }
50             printf(" characters are as a comment.*/");
51         }
52         for(i; i <= 61; i++){ /* decrypted other chars. */
53             int value = i - nyKey;
54             if(value < 0){
55                 value += 61;
56             }
57             if(alphabet[i] == tempChar){
58                 printf("%c", alphabet[value]);
59             }
60         }
61     }
62 }
63
64 }
65 return 0;
66 }
```

## DECRYPTED FILE(OUTPUT-2)

A screenshot of a text editor window titled 'text Editor'. The window has a menu bar with 'File' and 'Edit' options. Below the menu bar is a toolbar with 'Open' and 'Save' buttons. The file name 'decrypted.txt' and its path '~/Desktop' are shown in the title bar. The editor contains a C program with the following code:

```
1 #include <stdio.h>
2
3 int main() { /*There is a 22 characters are as a comment.*/
4     int number_one = 3;
5     int number_two = 4;
6     int add = number_one + number_two;
7     return 0;
8 }
```

**VIDEO – LINK :** (Sorry for bad quality.)

<https://youtu.be/4HwFwJnx7Yo>